

Generation Interconnection System Impact Study Report

For Queue Project Q20-1462

Carlin 138 kV

April 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1462. The Interconnection Customer proposes a 285.4 MW natural gas combined cycle facility interconnecting at the Carlin 138 kV substation in the Virginia area. The request was received on April 21, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 374 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,464,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Elko – Turner Rd 138 KV	Haggin – Mitchell Dam 345 KV	111.2	315	5.1
Redbed Plains Wind Farm Substation – Westcoast Cellufibre 138 KV	Trabue – Carlin 138 KV	104.5	380	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elko – Turner Rd 138 KV	Rebuild line with higher-capacity conductor	\$4,106,000	6% (\$233,000)
Redbed Plains Wind Farm Substation – Westcoast Cellufibre 138 KV	Reconductor line	\$2,091,000	59% (\$1,231,000)
	Total allocated to project		\$1,464,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.